

ALF basics: Overview

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ALF basics: Introduction

Lectures series on the programming details of working with ALF: ■

- Overview *This Talk*
- Building blocks of a Model
 - Lattice *Following Talk*
 - Observables *Tue. 13:00-13:30*
 - Interactions *Tue. 16:30-17:00*
 - Hopping *Wed. 13:00-13:30*
- Building your own Model/Hamiltonian *Wed. 16:30-17:30*
- Advanced pyALF *Thu. 13:00-13:30*

Folder overview: Miscellaneous

```

ALF/
├── Documentation/
│   └── doc.pdf ..... Documentation
├── Scripts_and_Parameters_files/ ..... Sample scripts
│   ├── Start/ ..... Files needed for ALF run
│   │   ├── parameters ..... Sample parameter file
│   │   ├── seeds ..... Random number seeds
│   │   └── out_to_in.sh ..... For continuing a run
├── testsuite/ ..... Suite of unit tests
└── Mathematica/ ..... Mathematica notebooks
                        for Wick decomposition

```

Folder overview: Monte Carlo code

ALF/	
— Analysis/.....	Error analysis programs
— HDF5/.....	HDF5 library, automatically downloaded and compiled
— Libraries/.....	General routines, e.g. Lattice
— Prog/.....	Monte Carlo core and Hamiltonians
— configure.sh.....	Sets environment variables
— Makefile.....	Needed by make

Folder Prog/

```

ALF/
├── Prog/ ..... Monte Carlo core and Hamiltonians
│   ├── Hamiltonians.list ..... Hamiltonian names
│   ├── Hamiltonian_main_mod.F90 .....
│   │   │   Interface between Hamiltonians
│   │   │   and Monte Carlo core
│   ├── Hamiltonians/
│   │   ├── Hamiltonian_MyModel_smod.F90
│   │   ├── Hamiltonian_MyModel_read_write_parameters.F90.
│   │   │   Automatically generated file
│   ├── main.F90 ..... Main program file
│   └── ALF.out ..... Executable

```